Sleep Disturbances and Risk of Sports Injury Among Collegiate Student-Athletes

Brett A. Messman, PhD,*§ Kyla A. Petrie, MD,† E. Whitney G. Moore, PhD,‡ and Trent A. Petrie, PhD*

Abstract

Objective: The purpose of this study was to examine the relationship between collegiate student-athletes' sleep and injury status. **Design:** This study was a secondary analysis of observational survey data of student-athletes collected across 2 time periods, October 2020 and January 2021. **Setting:** Data were collected as part of an ongoing national (USA) longitudinal survey investigating student-athletes' well-being in the aftermath of the COVID-19 pandemic. **Participants:** Participants were made-up of a subsample of participants from the parent study. Data from 89 exact matched pairs (across 5 demographics) of injured and noninjured student-athletes (68.5% NCAA Division I, 82.0% women) were assessed. **Independent Variables:** Student-athletes' sleep disturbance levels were the primary independent variable of interest. Analytic models also covaried for gender, racial, and ethnic identities, sport in-seasons, and previous injury status. **Main Outcome Measures:** Participant endorsed injury status was the main outcome measure of interest. Hypotheses were formulated during the data collection of the parent study. **Results:** Student-athletes injured in January 2021 reported higher average sleep disturbances in both October 2020 and January 2021 than noninjured student-athletes but not student-athletes injured in October. In addition, October 2020 sleep disturbances were a significant predictor ($P = 0.007$) of injury, such that odds of student-athletes being injured increased 1.07 times for each unit increase in their sleep disturbances controlling for their identities, sport in-season, and previous injury status. **Conclusions:** Findings signal support for the screening and targeting of student-athletes' sleep problems to reduce potential occurrences of sports-related injuries in student-athletes.

Key Words: student-athletes, sleep disturbances, sport, injury, NCAA

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INTRODUCTION

Collegiate athletic programs have begun to recognize that sleep is an important factor for student-athlete health, well-being, and performance.[1] Despite this emerging understanding, 39% to 61% of student-athletes regularly report experiencing some form of sleep disturbances[2–5] and are at higher risk of experiencing sleep disturbances than their nonathlete peers.[6–8] Student-athletes are particularly vulnerable to sleep disturbances due to intense physical training, sport performance pressures, and frequent travel across time zones.[2,4,9] For example, in 2015, NCAA student-athletes reported spending 27 to 41 h/wk on sport activities alone, while balancing academic and social obligations, sometimes at the detriment of nightly sleep.[1] Of particular concern is an emerging link between sleep disturbances and sports-related injuries,[6,10] which can affect the long-term health, well-being, and performance of student-athletes.

When considering the link between sleep disturbances and sports-related injuries, 1 line of research suggests that sports-related injury may predict subsequent experiences of sleep disturbances in student-athletes. Researchers have shown that sleep disturbances develop following various types of sports-related injuries; sleep disturbances are one of the most commonly reported complaints by athletes following a traumatic brain injury (TBI).[11] Previous evidence suggests that sleep disturbances can develop immediately following a TBI or emerge later during the recovery and persist for years after the initial TBI.[11] In addition, individuals who have sustained a traumatic spinal cord injury report greater sleep disturbances compared with noninjured comparison groups.[12–14] For example, Aydin and colleagues[14] reported that 74.7% of traumatic spinal cord injury patients experienced subsequent subjective sleep disturbances. Finally, DePhillipo et al[15] found that the preoperative incidence of sleep disturbances was 20% among arthroscopic knee surgery patients, whereas the postoperative incidence was 99%, 96%, and 90% at 1 week, 3 weeks, and 6 weeks, respectively. Together, such research indicates that sports-related injury incidence may be an important risk factor of developing student-athletes' short- and long-term sleep disturbances, which in turn may affect their long-term health, well-being, and performance.[1]

Conversely, a second line of research suggests that sleep disturbances may predict subsequent incidence of sports-

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*From the *Department of Psychology, University of North Texas, Denton, Texas; †School of Medicine, Texas Tech University Health Sciences Center, Lubbock, Texas; ‡Department of Kinesiology, College of Health & Human Performance, East Carolina University, Greenville, North Carolina; and §Center for Healthy Aging, The Pennsylvania State University, University Park, Pennsylvania.*

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Corresponding Author: Brett A. Messman, PhD, Center for Healthy Aging, The Pennsylvania State University, 201 Pollock Road, University Park, PA 16803 (bradamessy@gmail.com).

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related injury. Several studies provide that evidence sleep disturbances (eg, shorter sleep duration, poor sleep quality) are associated with greater risk of sports-related injury among student-athletes.[10,16–18] Haraldsdottir et al[16] found that collegiate volleyball players were more likely to sustain a sports-related injury following a night of below average sleep duration. Furthermore, sleep disturbances have been shown to negatively affect recovery following a sports-related injury, which in turn can exacerbate the injury, prolong the recovery period, and/or increase the risk of subsequent reinjury.[11,19] Critically, this line of research indicates that previous sleep disturbances may be an important risk factor for student-athletes' sports-related injury.

Taken together, findings indicate a need for further exploration into the relationships between sports-related injury and sleep disturbances to help identify student-athletes at most risk and inform potential intervention strategies. Critically, several gaps in the literature exist. First, with the exception of studies investigating the relationship between incidence of sports-related TBIs and sleep disturbances, there is limited research on the associations between general sports-related injury and sleep disturbances. Given that TBIs and spinal injuries are in the minority of reported sports-related injuries sustained by student-athletes,[10] this represents a potentially important, yet underreported research topic that requires further investigation. Second, the previous studies that have reported on the associations between incidence of general sports-related injury and sleep disturbances in student-athletes are limited by cross-sectional study designs and lacked an adequate noninjured comparison group.

The Current Study

To address these gaps, this study leveraged available data from a large, longitudinal national survey of NCAA student-athletes' health and well-being to examine the links between sports-related injury and sleep disturbances. Specifically, this study tested the association between incidence of sports-related injury predicting sleep disturbances and sleep disturbances predicting subsequent incidence of sports-related injury. We hypothesized that (1) student-athletes who reported having a sports-related injury would report greater sleep disturbances over time than a matched, noninjured sample[12–14] and (2) greater sleep disturbance levels would predict increased odds of reported subsequent sports-related injury.[10,16–18]

METHODS

Procedures

This study was a secondary analysis of a subset of data collected as part of a larger investigation on the longitudinal effects of COVID-19 on collegiate student-athletes' well-being (N = 2502). Participants were initially recruited into the parent study between April and May 2020 through an appendage to the end of the NCAA's Student-Athlete Advisory Committees study and solicitation of NCAA DI, DII, and DIII athletic departments.[20,21] Following consent, participants completed validated questionnaires online through Qualtrics. April/May participants were invited by email to complete subsequent follow-up surveys administered 3 months after the previous survey. Starting with the October 2020 follow-up survey, questions regarding sports-related injury incidence were included (ie, October 2020 and January 2021 follow-up surveys) because student-athletes returned to sport training and competitions after collegiate sports previously being cancelled due to COVID-19. Leveraging the parent study's longitudinal data collection, we set-up a 2 timepoint naturalistic experimental design around the reported incidence of sports-related injuries at the October 2020 and January 2021 follow-up surveys.

Subjects

Participants were selected for this study in 2 steps. First, we identified student-athletes who had reported a sports-related injury in the January 2021 follow-up survey (n = 89; ie, ~4% of surveyed sample, which is comparable to the proportion of injured student-athletes sampled in other similar prospective studies[22]). Of these, 48 student-athletes reported being injured in October 2020 and still injured in January 2021, whereas the other 41 student-athletes were reporting new injuries in January 2021. Second, noninjured student-athletes who exactly matched the injured athletes on all 5 (5) characteristics – NCAA Division (I, II, or III), sport, gender identity, race identity, and Hispanic identity – comprised the comparison sample. We identified exact matches across all 5 demographics for 83 of the 89 injured student-athletes. The remaining 6 injured student-athletes reported mixed racial identity or a racial identity that was unique among our participants of the same NCAA Division, sport, gender, and Hispanic identities. We were able to exactly match these 6 student-athletes with noninjured student-athletes across the other 4 categories and the matches were not of the majority race (ie, White). Thus, the final dataset was comprised of 89 matched pairs of injured and noninjured student-athletes (178 individuals). Of the 89 participant pairs, 73 identified as women; 74 identified as White; 83 identified as Not Hispanic; and 61 identified as NCAA Division I student-athletes. Given this low representation of individual minority status, for the subsequent analyses, individuals were grouped as Athletes of Color and White (reference group). See Tables 1 and 2 for details.

Measures

Demographics

The participants reported on their demographics including NCAA Division (I, II, III), sport (eg, football, field hockey, baseball, softball, track and field), gender identity, racial identity, and Hispanic identity.

Injury Status

Participants responded if they were currently injured (yes/no). If yes, they were presented with open-ended questions to describe the injury type and number of days missed in their sport participation due to being injured.

Sleep Disturbances

The student-athletes' sleep disturbances over the past 4 weeks were measured with the Patient-Reported Outcomes Measurement Information System Sleep Disturbance Short Form (PROMIS-SD).[23] The sleep items were summed and that sum

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TABLE 1. Student-Athlete Demographics (N = 178)

Sleep Disturbance Grouping	Injured n (%)		Not Injured n (%)	
None to slight				
October 2020	48	(53.9)	63	(70.8)
January 2021	51	(57.3)	62	(69.7)
Mild				
October 2020	21	(23.6)	14	(15.7)
January 2021	20	(22.5)	16	(18.0)
Moderate				
October 2020	17	(19.1)	11	(12.4)
January 2021	17	(19.1)	10	(11.2)
Severe				
October 2020	3	(3.4)	1	(1.1)
January 2021	1	(1.1)	1	(1.1)
Ethnicity				
Hispanic	6	(6.7)	4	(4.5)
Not Hispanic	83	(93.3)	85	(95.5)
Race (non-Hispanic n = 3521)				
Caucasian/White	74	(83.1)	74	(83.1)
African American/Black	3	(3.4)	2	(2.2)
Asian/Asian American/Pacific Islander	5	(5.6)	6	(6.7)
Multiracial	6	(6.7)	5	(5.6)
Other	1	(1.1)	2	(2.2)
Academic year				
1st year	12	(13.5)	17	(19.1)
2nd year	25	(28.1)	23	(25.8)
3rd year	30	(33.7)	35	(39.3)
4th year	7	(7.9)	11	(12.4)
5th year	2	(2.2)	3	(3.4)
Not report	13	(14.6)		
Gender identity				
Women	73	(82.0)	73	(82.0)
Men	16	(18)	16	(18)
NCAA Division				
Division I	61	(68.5)	61	(68.5)
Division II	12	(13.5)	12	(13.5)
Division III	16	(18.0)	16	(18.0)
Sport participation				
Baseball	1	(1.1)	1	(1.1)
Basketball	3	(3.4)	3	(3.4)
Cross-country	22	(24.7)	22	(24.7)
Field hockey	3	(3.4)	3	(3.4)
Football	1	(1.1)	1	(1.1)
Golf	1	(1.1)	1	(1.1)
Gymnastics	9	(10.1)	9	(10.1)
Lacrosse	2	(2.2)	2	(2.2)
Rowing	3	(3.4)	3	(3.4)
Soccer	12	(13.5)	12	(13.5)
Softball	3	(3.4)	3	(3.4)
Swimming & diving	9	(10.1)	9	(10.1)
Tennis	4	(4.5)	4	(4.5)
Track and field	10	(11.2)	10	(11.2)
Volleyball	4	(4.5)	4	(4.5)
Wrestling	1	(1.1)	1	(1.1)
Equestrian	1	(1.1)	1	(1.1)

score converted to a T-score,[23,24] so the values could be interpreted regarding established norms for severity: <55 = none to slight; 55.0 to 59.9 = mild; 60.0 to 69.9 = moderate; and ≥70 = severe. The sleep disturbance scores had acceptable reliability ($\alpha_{October2022} = 0.92$; $\alpha_{January2021} = 0.92$).

Analysis Plan

Data Preparation

The data at each timepoint passed univariate normality tests (skew and kurtosis values ≤ |3|; Tabachnick & Fidell, 2007). Of the 178 participants, all provided responses at both timepoints.

Analysis of Sleep Disturbances Over Time x Injury Status

To test how student-athletes' sleep disturbance levels differed over time by injury status (Hypothesis 1), we conducted a 2 × 3 repeated measures ANOVA (rmANOVA) over 2 time periods (October 2020 and January 2021) and 3 groups (injured in October, injured in January, and not injured). Although participant pairs were matched on demographics, individuals' demographics could influence their sleep disturbances; therefore, we included covariates representing sport in-season (ie, Fall and Winter, with Spring as the reference season), as well as gender identity (female as the reference group), race, and ethnicity identities (White as the reference group). Significant results were probed in follow-up analyses and graphically presented with 95% CIs about the mean values included. Cohen's *d* effect sizes with pooled SD (SD) were calculated over time (within group) and between groups (within time).

TABLE 2. Sport Injury Categories for October 2020 and January 2021 Injuries

Injury Type	Injured n (%)	
Head/TBI/Back/Spine		
October 2020	6	(53.9)
January 2021	7	(57.3)
Orthopaedic surgical		
October 2020	19	(23.6)
January 2021	6	(22.5)
Orthopaedic nonsurgical		
October 2020	19	(19.1)
January 2021	22	(19.1)
Other (e.g., illness, vague)		
October 2020	4	(3.4)
January 2021	6	(1.1)

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TABLE 3. Sleep Mean Scores, Standard Deviations, & Cohens d for Injury Status Groups by Time

	Injury Grouping	Mean	SD	1a	1b	1c	2a	2b
October	1a. Noninjured Match	50.47	8.38					
	1b. Injured in January	54.77	8.47	0.51				
	1c. Injured in October	53.18	7.92	0.33	0.33			
January	2a. Non-injured Match	50.46	8.06	0.00	−0.53	−0.34		
	2b. Injured in January	54.41	8.23	0.47	−0.04	0.15	0.49	
	2c. Injured in October	51.18	8.35	0.09	−0.43	−0.25	0.09	−0.39

October represents October 2020; January represents January 2021. Those injured in January had significantly ($P < .05$) higher sleep disturbance scores in October and January compared with the other 2 injury groups.

Analysis of Sleep Disturbance Levels Predictive of Injury

To test the effect of previous sleep disturbances on individuals' odds of being injured in the future (Hypothesis 2), we conducted a logistic regression with the matched samples. Student-athletes' October sleep disturbance scores were included as the predictor after controlling for their sport in-season and identities (ie, gender, race, and ethnicity) and the dependent variable was coded 0 (not injured in October or January) and 1 (injured in January).

Ethical Considerations

All study procedures were approved by the The University of North Texas Institutional Review Board.

RESULTS

The frequency of sleep disturbances was examined at each timepoint. Based on clinical cut-off scores of the PROMIS-SD, the majority (51%-71%) of the participants at each timepoint were classified as experiencing none to slight sleep disturbances, followed by fewer experiencing mild (16%-29%), moderate (11%-19%), and severe sleep disturbances (1%-3%).

Sleep Disturbances over Time by Injury Status (Hypothesis 1)

The data met the assumptions of equality of covariances by a nonsignificant Box's Test ($P = 0.815$), sphericity (Mauchly's $W = 1.00$, $P = 1.00$), and homogeneity of variances by nonsignificant Levene's tests for the October 2020 ($F_{(1, 175)} = 0.68$, $P = 0.510$) and January 2021 ($F_{(1, 175)} = 0.12$, $P = 0.890$) sleep data. Based on the rmANOVA, none of the covariates were significant: Fall in-season ($F_{(1, 170)} = 2.39$, $P = 0.124$), Winter in-season ($F_{(1, 170)} = 3.54$, $P = 0.060$), Hispanic ($F_{(1, 170)} = 0.13$, $P = 0.720$), Nonwhite ($F_{(1, 170)} = 0.002$, $P = 0.960$), and male ($F_{(1, 170)} = 3.06$, $P = 0.080$). There was a significant difference in sleep scores by injury status ($F_{(2, 170)} = 4.53$, $P = 0.012$) and accounted for 5.1% of the sleep score variance (partial $\eta^2 = 0.051$; Table 3 and Figure 1). Based on the Bonferroni corrected pairwise comparisons (Table 4), for the October 2020 sleep disturbance scores, the student-athletes injured in January ($M = 54.77$, 95% CI, 52.19-57.26) reported higher sleep disturbances than noninjured student-athletes ($M = 50.47$, 95% CI, 48.76-52.20) but not student-athletes who were injured in October ($M = 53.18$, 95% CI, 50.84-55.55). In addition, for January 2021 sleep disturbance scores, the student-athletes injured in January ($M = 54.41$, 95% CI, 51.79-56.86) reported sleep disturbances remained significantly higher than the noninjured student-athletes ($M = 50.46$, 95% CI, 48.70-52.14), but not the student-athletes injured in October ($M = 51.18$, 95% CI, 48.97-53.68).

Sleep Disturbances Predicting Odds of Future Injury Status (Hypothesis 2)

The logistic regression predicting student-athletes' injury status in January 2021 followed a stepwise approach. The null model without any predictors had poor classification (50% correctly classified; no better than chance). Next, the

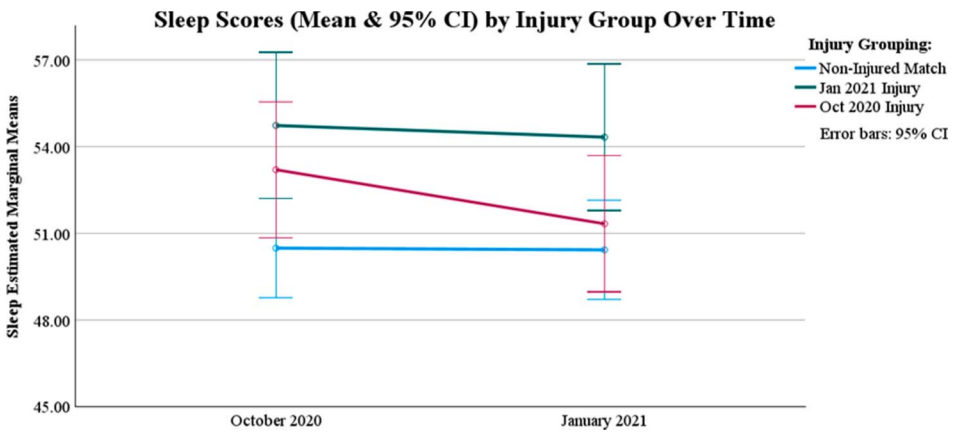

Figure 1. Graph of collegiate student-athletes' sleep disturbance by injury group.

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TABLE 4. Bonferroni Corrected Pairwise Comparisons

	Grouping I	Grouping J	Mean Difference (I-J)	*P*
October	Non-injured	Injured in January	−4.24	**0.020**
		Injured in October	−2.71	0.206
	Injured in January	Non-injured	4.24	**0.020**
		Injured in October	1.53	0.999
	Injured in October	Non-injured	2.71	0.206
		Injured in January	−1.53	0.999
January	Non-injured	Injured in January	−3.90	**0.038**
		Injured in October	−0.90	0.999
	Injured in October	Non-injured	3.90	**0.038**
		Injured in January	3.00	0.272
	Injured in January	Non-injured	0.90	0.999
		Injured in October	−3.00	0.272

Bolded values represent statistical significance, $P < 0.05$.
October represents October 2020; January represents January 2021.

model with categorical covariates (gender, race, Hispanic), sport in-season (Spring reference category), and October 2020 injury status fit significantly better than the null model ($\chi^2 = 85.275$, $df = 6$, $P < 0.001$), accounted for 51% of the injury status variance (Nagelkerke pseudo-R^2), and resulted in the correct classification of 77% of student-athletes' January 2021 injury status. All noninjured student-athletes were correctly classified; however, 41 of the injured student-athletes were misclassified by this model. Adding the student-athletes' October 2020 sleep disturbance scores to the model significantly improved fit ($\chi^2 = 8.08$, $df = 1$, $P = 0.004$), accounted for a total of 54% of the injury status variance, and was slightly better with athlete classification (78%). This model correctly classified 87 of the 89 noninjured student-athletes and 51 of the 89 injured student-athletes (Table 5). The October 2020 sleep disturbance score was a significant predictor ($P = 0.007$) such that the odds of student-athletes being injured increased 1.07 times for each unit increase in their sleep disturbance T-scores, controlling for their identities, sport in-seasons, and previous injury status

TABLE 5. Output About Classification From the Final Logistic Regression Model

	Predicted		
Observed	Not Injured	Injured in January	Percentage Correct
Not injured	87	2	97.8
Injured in January	38	51	57.3
Overall Percentage			77.5

January represents January 2021.

(Table 6). In practical terms, compared with the 13% probability of injury for student-athletes' reporting slight sleep disturbances (scores < 55), student-athletes' injury probability doubled to 26% with mild sleep disturbances (scores 55-59.9), 28% with moderate sleep disturbances (scores 60-69.9), and 33% with severe (scores ≥70) sleep disturbances.

DISCUSSION

We examined associations between sleep disturbances and sports-related injury in student-athletes, generally replicating and extending findings from previous studies.[10,16–18] Results indicated that heightened sleep disturbances were associated with subsequent incidence of sports-related injury in student-athletes. In addition, we showed that participants' injury probabilities increased with the severity of their sleep disturbances. Findings support the view that maintaining healthy sleep in student-athletes may be an important preventative measure for reducing risk of sports-related injury.[1]

Partially consistent with Hypothesis 1, student-athletes injured in January 2021 had greater sleep disturbances before and concurrent with their reported incidence of sports-related injury compared with noninjured student-athletes. This result replicates previous studies showing that sleep disturbances may be elevated before a sustained sports-related injury.[10,16–18] However, for student-athletes injured in October 2020, sleep disturbances concurrent with and after their reported incidence of sports-related injury were not significantly different than either the injured in January 2021 or noninjured groups. Notably, there was a trend of sleep disturbances decreasing over time in this group in the aftermath of the reported sports-related injury. This finding is inconsistent with several previous studies that have shown individuals report greater sleep disturbances following a sustained injury.[11–15] However, these studies were generally focused on incidence of less commonly occurring specific injury types (eg, TBI or spinal) and may not be representative of the experiences of student-athletes who experience more commonly occurring injuries (eg, musculoskeletal or orthopaedic). Thus, this unexpected result may warrant a more nuanced investigation into how incidence of sports-related injury predicts subsequent sleep disturbances. Future studies may benefit from investigating factors, such as when the injury occurred,[11,15] type and severity of injury,[6] access to treatment,[25] and recovery behaviors (eg, treatment adherence),[26] to better understand who is at most risk for developing long-term sleep disturbances following a sustained sports-related injury.

Consistent with Hypothesis 2, our results indicated that greater sleep disturbances in October 2020 were associated with increased odds of a reported sports-related injury in January 2021. Specifically, January injury probability for participants classified as having clinically mild (26%; scores 55-59.9), moderate (28%; scores 60-69.9), or severe (33%; scores ≥ 70) sleep disturbances in October 2020 was over double that of participants who were classified as having none to slight sleep disturbances (13%; scores <55). This result corroborates and extends our previous result and evidence from previous studies by indicating that sleep disturbances may be a risk factor of future incidence of sports-related injury in student-athletes.[10,16–18] Critically, our findings

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TABLE 6. Model 2 Regression Coefficient Output

Predictors	*b*	*SE*	*P*	OR	OR 95% CI
Intercept	−4.75	1.57	**0.002**		
Fall season	0.28	0.51	0.581	1.33	[0.49-3.62]
Winter season	0.25	0.61	0.685	1.28	[0.39-4.23]
Ethnicity	−1.07	1.25	0.389	0.34	[0.03-3.93]
Race	0.26	0.54	0.637	1.29	[0.45-3.71]
Gender	0.34	0.54	0.534	1.40	[0.48-4.08]
Injured in October	22.22	5610.75	0.997	4 444 020 661.50	0, INF
Sleep disturbances in October	0.07	0.03	**0.007**	1.07	[1.02-1.13]

Bolded values represent statistical significance, $P < 0.05$.
95% CI, the 95% confidence interval of the odds ratio; b, unstandardized beta estimate; OR, Odds Ratio; SE, standard error.

demonstrate student-athletes with clinically moderate-to-severe sleep disturbances are at a greatest risk of future incidence of sports-related injury.

Study Limitations and Future Directions

Although the study had several strengths (ie, 2 timepoint longitudinal study design and matched injured and noninjured groups), there are also limitations worth noting. First, while matching the sample of injured student-athletes to a subset of noninjured student-athletes aided in providing a more adequate comparison group, this study's participants were primarily female, Caucasian/White, and Non-Hispanic, which is not reflective of the larger USA student-athlete population, and thus limits the generalizability of study findings. Previous studies have shown important racial/ethnic and gender-based disparities in sleep[27,28] and sports injury incidence and recovery.[29,30] In addition, generalizability of study findings were limited by the timing of data collection, which did not include in-season for Spring sports. Future research should aim to replicate results across more diverse samples and across all sports in-seasons to enhance generalizability.

Second, although we used a validated measure of subjective sleep disturbances with established clinical categories of sleep disturbance severity norms, there are limitations to using self-report clinical cut-off scores (vs administered clinical assessments).[31] Furthermore, student-athletes were drawn from a nonclinical population; as with previous studies, the majority were categorized with either none or mild sleep disturbance severity at both timepoints. Thus, our results may underestimate the true effect of more clinically severe levels of sleep disturbances (PROMIS-SD T-Scores $\geq$ 60) on sports injury incidence. Future studies should try to replicate these findings using more rigorous clinical assessment. Likewise, our assessment injury status was also self-reported by participants at both time points, and future studies should seek to confirm sustained sports injuries through clinical assessment or verification of a diagnosis for a more valid assessment of injury. In addition, although we had some information on injury type, injury type groupings were too small to meaningfully explore differences in the associations between sleep disturbances and incidence of sports-related injury by this variable. Future inquiries would benefit from a more nuanced comparison of the sleep-sports-related injury by injury type.[6]

Finally, the large time gap (ie, 3 months) between the timepoints may have been a potential reason for the smaller than expected observed effects. Despite this, results do support future investigation into associations between sleep disturbances and sports injury. As a next step, researchers could use study findings to design intensive longitudinal study designs to investigate more nuanced temporal dynamics between sleep disturbances and the incidence and recovery of sports injuries in student-athletes.

CONCLUSIONS

According to a recent consensus statement from the NCAA Interassociation Task Force on Sleep and Wellness, healthy sleep is critical for student-athletes' health, well-being, and performance.[1] Along with findings from other studies,[10,16–18] our work shows that sleep disturbances may be a risk factor of sustained sports-related injury in student-athletes. We encourage researchers to consider exploring the unique associations of sleep disturbances and incidence of sports-related injury by injury type in more diverse and clinical samples. Despite study limitations, results signal support for screening and targeting student-athletes' sleep problems to reduce potential risk of sports-related injuries. Consistent with NCAA best practices (2024),[32] we encourage collegiate athletic departments to consider periodic monitoring student-athletes' sleep health and treatment referrals for sleep-related complaints to reduce injury risk.

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